

PROJECT 4 — SHEET-METAL INSTRUMENT ENCLOSURE

Project Type: CAD / 3D Modelling Capstone

Client Brief

Design a sheet-metal enclosure for an electronic instrumentation module. The enclosure must accommodate a PCB, connectors, controls, mounting points, ventilation and a removable cover, and must produce a manufacturable flat pattern.

Design Specification

- External envelope: approximately $250 \times 180 \times 100$ mm
- Material: 1.5 mm sheet aluminium
- PCB envelope: approximately 200×130 mm
- Front: display, two connectors and switch
- Rear: power and communication connectors
- Bottom: four mounting points
- Top: removable cover

Parametric Requirements

SHEET_THICKNESS, BEND_RADIUS, K_FACTOR, ENCLOSURE_LENGTH, ENCLOSURE_WIDTH, ENCLOSURE_HEIGHT, FLANGE_LENGTHS, HOLE_LOCATIONS and PCB interface dimensions.

Required Skills

- Sheet-metal modelling
- Flanges and bends
- Bend radius and bend allowance
- K-factor / bend deduction concepts
- Bend and corner relief
- Unfold/fold
- Flat pattern generation
- DXF preparation
- Manufacturing drawings
- DFM review
- Flat pattern → 3D and 3D → flat pattern

Client Approval Questions

- What sheet thickness and bend radius are assumed?
- Is the bend sequence physically feasible?
- Are flange widths manufacturable?
- Are holes sufficiently clear of bends?
- Are corner/bend reliefs adequate?

- Can the flat pattern be cut and folded without collisions?
- Does the folded model reproduce the intended enclosure?

CAD QA / Validation

- Flat-pattern validity
- Bend collision review
- Hole-to-bend clearance
- Material/thickness consistency
- DXF geometry check
- Drawing/model consistency
- Manufacturing feasibility review

Required Deliverables

- Native FreeCAD model (.FCStd)
- Manufacturing / engineering drawing (PDF)
- STEP export
- STL export where relevant
- Parameter/design-intent record
- CAD QA checklist/report